Supporting Information: Statistical Assessment of the influence of Morphological and Compositional Variability over Electronic Properties in Palladium Dodecanethiolate Nanoparticles

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Contents

1 Data archive **S-3**

2 Distribution Fitting **S-3**

3 X-Ray Photoelectron Spectroscopy **S-3**

4 Continuous Wave Electron Spin Resonance Fitting **S-3**

5 Correlations **S-6**

1 Data archive

All data used in this manuscript, Python code used to process the data, images used in the manuscript, and Veusz files can be obtained at the Penn State data commons archive at: *link to be added upon acceptance.*

2 Distribution Fitting

Fitted distributions for nanoparticle size, aspect ratio, and circularity were performed following our previous work in: https://pubs.acs.org/doi/10.1021/acsnanoscienceau.4c00076. Details of the fitting procedure can be found in the SI of that publication.

3 X-Ray Photoelectron Spectroscopy

X-ray photoelectron spectroscopy (XPS) was obtained at the Pennsylvania State University Materials Characterization Lab Core Facility, Materials Research Institute. Fitting of peaks and compositional atomic percents (%) were performed by Bob Hengstebeck on a Physical Electronics VersaProbe III instrument equipped with a monochromatic Al kα X-ray source (hν = 1,486.6 eV) and a concentric hemispherical analyzer.

4 Continuous Wave Electron Spin Resonance Fitting

Fitting was done using the `lmfit` library of Python. To determine the appropriate fit, wide-scan CW-ESR spectra were collected using a Bruker ELEXSYS E-500 spectrometer (Bruker BioSpin Corporation, Billerica, MA), with the corresponding X-band bridge and an ELEXSYS Super High Sensitivity cavity. An ESR 900 liquid helium (LHe) cooled top-loading continuous flow cryostat was controlled by an Oxford Instruments ITC502 temperature controller. All measurements were taken with a modulation amplitude of 1 mT, 6.325 mW

microwave power, and a 500 second scan time over a range of 600 mT. We note the absence of R1's spectrum due to the sample being lost in the one year time frame. Additionally, while the secondary characterization was performed a year after synthesis, the samples were stored in capped EPR tubes at liquid nitrogen temperatures. Previous studies by *Fagan et. al.* support minimal modulation of electronic properties when the samples are frozen for extended periods of time.[?] Thus, while we anticipate differences in the isotropic g-value, these should be minimal. Rather than focusing on the variation in this value, the purpose of illustrating this data is to exemplify how a Voigt powder pattern was decided as the appropriate fitting method. Isolated g-values can be found in Table S1 with the corresponding fits in Figure S1.

Table S1: Anisotropic and isotropic g-values for PdSC-12 NPs.

	g_x	g_y	g_z	g_{iso}
R1	—	—	—	—
R2	1.8227 (0.00090)	2.0139 (0.00059)	2.0628 (0.00035)	1.9665 (0.0004)
R3	1.8953 (0.00047)	2.0239 (0.00048)	2.0666 (0.00026)	1.9953 (0.0002)
R4	1.8990 (0.00063)	2.0359 (0.00066)	2.0663 (0.00048)	2.0004 (0.0003)
R5	1.8997 (0.00172)	2.0511 (0.00229)	2.0673 (0.00216)	2.0060 (0.0012)

Determination of the lineshape was a continuation of *Cruz et. al..*[?] For this work, we examined the models proposed in that work. We also considered powder patterns based on pure Gaussian, pure Lorentzian, and Voight profiles. We created models that also accounted for the polydispersity in size. These models are summarized in Figure S2. We found that the lineshapes shown in Figure S2 were best fit by a Voigt-based powder pattern. The generalized form is:

$$V_p(B) = \frac{d}{dB}\left[I\sum_{i=1}^{N} V(B - B_i; \sigma_i, \gamma_i)\right] \tag{1}$$

Where $V_p(B)$ is the simulated Voigt powder pattern of the ESR signal as a function of magnetic field, I is an intensity scaling factor, B is the applied magnetic field, N is the number of sampled orientations in the powder average, $V(B-B_i;\sigma_i,\gamma_i)$ is the Voigt lineshape centered at the orientation-dependent resonance field with Gaussian σ_i and Lorentzian γ_i

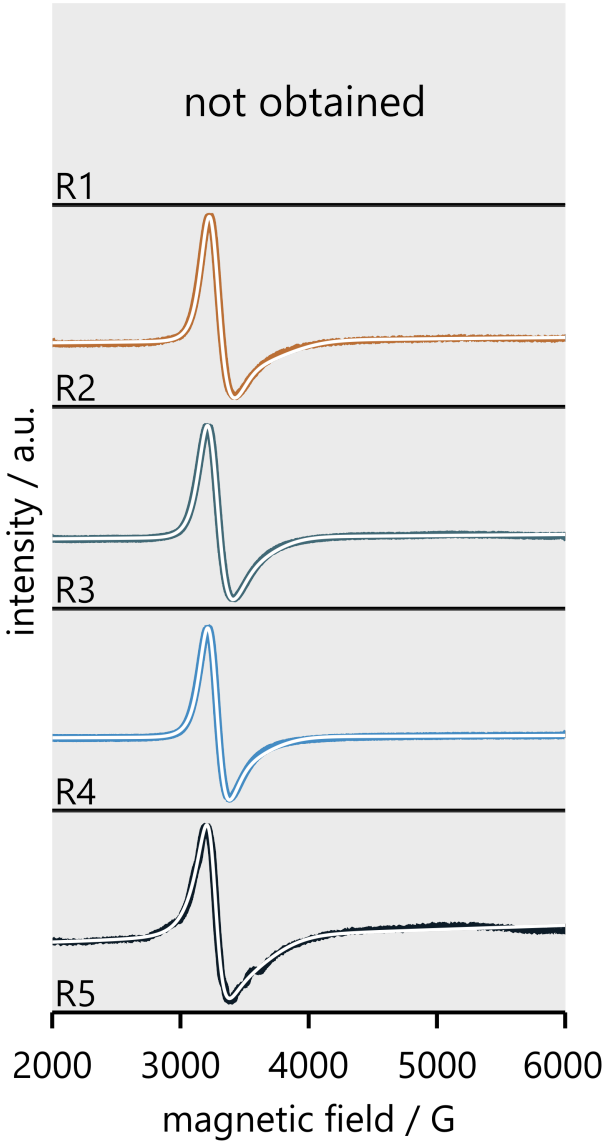

Figure S1: CW-ESR spectra obtained for each of Pd-SC12 NPs. Spectra ranged from 400-6400 Gauss to fully capture the baseline and avoid truncation, while Voigt powder fitting was performed from 2000-6000 Gauss. Experimental spectra are shown in color and fits in white.

broadening, and B_i is the resonance field for the i^{ith} orientation. This model was selected as the major contribution to signal asymmetry is believed to arise from the heterogeneous nanoparticle morphology in different chemical environments.

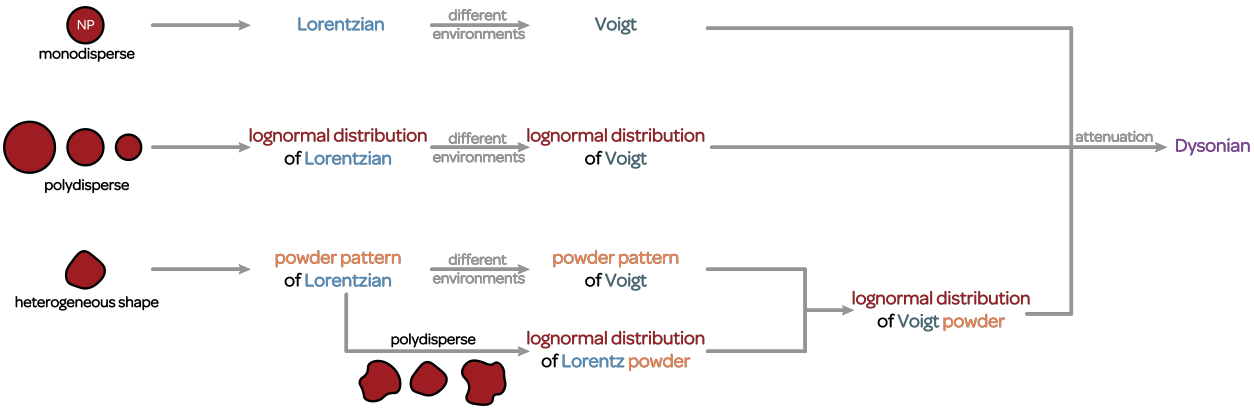

Figure S2: Possible lineshapes for asymmetric CESR spectra of metallic nanoparticles.

Similarly, a Welch's t-test was used to evaluate if the original g-values were significantly different from the new g-values as seen in Figure **??**. This analysis revealed that R1a remained not significantly different from both extracted R3 values. Interestingly, R1a was also not significantly different from R4b and R5b. However, in the main paper, given the original measurements were made on the same day, these pairs were different. Similar to the discussions in the main paper, we would expect measurements of the same sample would result in not significantly different values. However, all but one pairing (R5a/R5b), were significantly different. Given we discuss sources for this variation in the main text, we will not reiterate that discussion here.

5 Correlations

Beyond testing significant and equivalence, correlations between variables were analyzed using an analysis of variance via the `statsmodel` package. In this, we evaluated the mean diameter, aspect ratio, circularity, S:Pd molar ratios, Pauli paramagnetic susceptibilities, XPS sulfur ratios, and g-values. A correlation matrix (Figure S3A) was constructed that

highlights whether two variables are related.

While looking at Figure S3A, some variables such as S:Pd and χ^{Pauli} appear to be correlated. However, we note that the mathematics behind the calculation of χ^{Pauli} ensures some correlation and we are unable to judge if there is additional correlation beyond that expected from the calculation. Similarly, duplicate measurements tend to higher correlation, as expected. Interestingly, the second measurement of the g-values results in a higher correlation among all variables. However, this dependence is attributed to the reduced number of samples (four versus the five for all other measurements). From these results we conclude that the electronic properties are minimally dependent on the morphology and size at this magnitude of variation.

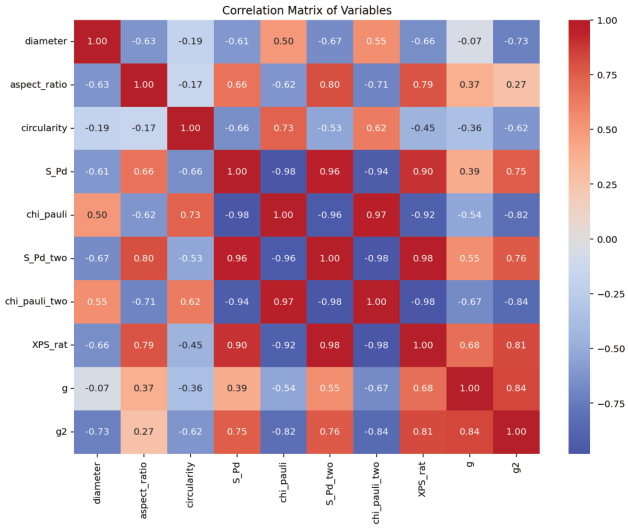

Figure S3: Correlation matrix for all variables tested. Matrix values nearer ±1 indicate high correlation.